

ATHLETEX TESTING REPORT

Execution Date: September 27, 2026 | Status: BUILD SUCCESS (100% Passed)

1. Executive Summary Table

Category	Tech Stack	Test Classes	Tests	Passed	Failed	Status
Backend Unit Tests	JUnit 5, Mockito	Auth, Badge, CoachReport Services	11	11	0	100% PASSED
REST Controller Tests	MockMvc, JUnit 5	AuthController, NotificationController	5	5	0	100% PASSED
Selenium E2E Tests	Selenium 4, Chrome	AthleteXWebE2ETest	2	2	0	100% PASSED
Frontend Component Suite	Vitest, RTL, jsdom	ProtectedRoute.test.jsx, Login.test.jsx	5	5	0	100% PASSED
TOTAL		23 Tests Executed	23	23	0	100% SUCCESS

2. Detailed Test Coverage Breakdown

Feature Area	Verification Scope & Assertions
Authentication & Security	Tested password complexity rules, duplicate email checks, login BCrypt matching, activity logs, & JWT issuance.
Badge Evaluation Engine	Unlocked First Step, Gold Standard, Coach Verified Champion, & Evidence Master badges based on criteria.
Coach Reporting Engine	Verified Individual Athlete Report metrics (Overall, Speed, Strength, Endurance, Agility) & unassigned coach security isolation.
Selenium E2E Automation	Headless ChromeDriver launched, navigated to http://localhost:5173, verified DOM #root & login forms.
Frontend UI & Routing	Verified ProtectedRoute unauthenticated redirects & role protection, and Login form bindings & submission.

3. Commands to Re-run Test Suite

```
# Backend Unit & Controller Tests:
.\mvnw.cmd test "-Dtest=AuthServiceTest,BadgeServiceTest,CoachReportServiceTest,NotificationControllerTest,

# Selenium E2E Browser Automation Tests:
.\mvnw.cmd test "-Dtest=AthleteXWebE2ETest"

# Frontend Vitest Suite: cd athletex-frontend && npm test
```